

Low Voltage Bandgap Reference Circuit with 13.14ppm/°C Using a Subthreshold Operational Amplifier

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Abstract— A low-power bandgap reference is presented that is able to function at a supply voltage of 1.2 V and a supply current of 74 ± 4 mA. The bandgap reference has a temperature coefficient of ± 7.2 ppm/°C. The reference voltage is 691 mV. The temperature range of the circuit is 0° to 70° C. The proposed bandgap voltage reference employs a low voltage CMOS bandgap core, PTAT IBias generator, op-amp, and a start-up circuit. The paper focuses on circuit analysis using Cadence with SPECTRE, Monte Carlo, and layout design.

I. INTRODUCTION

Bandgap voltage references are a widely used and fundamental component in many integrated circuits, as they provide stable and accurate reference voltages which are important for proper function of analog and mixed-signal systems. Bandgap voltage references have become an important piece in the rising technologies of battery-powered electronics for example. Many of these systems need stable references with low power consumption and wide operating ranges.

The approach chosen for our bandgap voltage reference involves the integration of a low voltage CMOS bandgap core with an op-amp operating the subthreshold region. The subthreshold op-amp was chosen as this provides incredibly high gain along with low power consumption. The op-amp is powered with a PTAT Bias generator which further contributes to temperature stability and precision. Our bandgap voltage reference includes start-up circuitry to enhance the transient response of our circuit for stability.

The next sections of the paper examine the intricate details of our circuit design and its performance. This will cover characteristics and demonstrate the viability for our low-voltage, low-power bandgap voltage reference design where a stable and accurate voltage reference is required.

II. CIRCUIT DESCRIPTION

This circuit is a bandgap voltage reference circuit. These circuits are widely used and fundamental circuits used to output a stable voltage value that remains stable and constant despite variations in process, voltage, and temperature. The circuit intakes a supply voltage that may be unstable and outputs a stable voltage output. Typically, this architecture tends to function with 3.3V supplies outputting at 1.2V which is close to

the bandgap Silicon but our design is able to function with only 1.2V supply voltage and less than 100uA of current draw.

This circuit works by producing two component voltages: One voltage component is proportional to changes in absolute temperature (PTAT) which is created by running two different diode connected transistors at different current densities. The other component voltage is complementary to absolute temperature (CTAT) which is created by running constant current through a diode. When the two components are properly scaled to each other and added, the resulting output voltage from both of the temperature dependent components cancel each other out and exhibit little to no change with temperature fluctuations.

This self-correction is fulfilled by a system of feedback with a low power, subthreshold op-amp that keeps the nodes and currents at their operational values by changing the biasing of a current mirror. This op amp itself needs to be biased which is done by a PTAT bias generator which itself delivers quasi constant current due to its exceptionally weak temperature dependence. There is however, a state of this circuit architecture where no current flows and all components remain stable with no perturbation. This circuit is meant to be self-stabilizing but needs a start-up circuit to ensure that it falls into the operational stability point and not the zero current stability point.

III. CIRCUIT ANALYSIS

A. Bandgap Core

To create the stable voltage reference a bandgap core is utilized. The bandgap core used here is adapted from Razavi[1]. The basic idea of a bandgap reference core is that since V_{BE} will decrease linearly with temperature and V_T increases linearly with temperature and those voltages can be scaled and summed to create V_{REF} . The equation is $V_{REF} = V_{CTAT} + V_{PTAT}$, Where CTAT is complementary to absolute temperature and PTAT is proportional to absolute temperature. In our case, the voltage across a base-emitter BJT is summed with a thermal voltage generator that is scaled by some temperature-independent scalar 'K': $V_{REF} = V_{BE} + K \cdot V_T$. Our V_{BE} being the V_{PTAT} and our 'K' and V_T are the V_{CTAT} .

To implement this idea by using PFET current mirrors formed by M_2 , M_3 , and M_4 . These current mirrors force currents I_1 and I_2 along each branch. The op-amp is used at the gates of the PFETs M_2 and M_3 to provide feedback to ensure the current is stable across temperatures. I_1 is the current formed through

$D_{7:1}$, D_0 and R_3 . I_1 is a PTAT current given by $I_1 = V_T \ln(\frac{N}{R_3})$. N is the number of diodes used for $D_{7:1}$, which is 7 diodes. Our CTAT current is formed by the voltage across the BJTs and R_2 and R_4 , which should have the same resistance, but we have found better results with these slightly different. We believe this comes from balancing the voltage drop across V_{BE0} to better match the drop across R_2 . I_2 , the CTAT current, has the equation $I_2 = \frac{V_{BE0}}{R_2}$. So V_{REF} is given by $(I_1 + I_2) * R_5$. The full equation is $V_{REF} = \frac{R_5}{R_2} * (V_{BE0} + (\frac{R_2}{R_3} * \ln(N) * V_T))$.

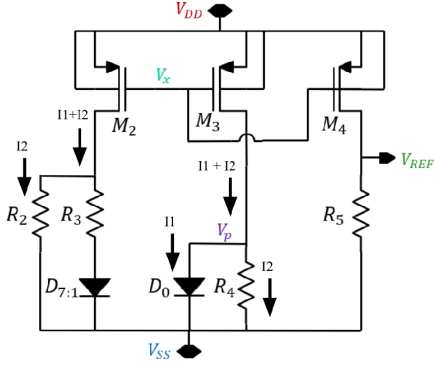


Figure 1: Schematic of the bandgap core, adapted from [1].

The group decided on resistor resistance by balancing current usage versus layout size. It was preferable to use large resistors because it would lower the current of the circuit, but it was important for the resistors to not be too large as it would take up a large footprint in layout.

To decide on the current mirror MOSFET sizing there was a need to increase the size from the original sizing to be more resilient against component mismatch in our Monte Carlo simulations. The sizing also had a strong effect on our V_{REF} so sizes were chosen that would match our calculations for the resistors and number of diodes. There also had to be caution as large sizes mean less space in layout. The sizing of M_2 and M_3 had to be the same to ensure similar currents flowing into both branches controlled by the op-amp.

After completing the calculation and adjusting the sizing of our current mirror of M_2 , M_3 , and M_4 a V_{REF} of 676mV with around a 1mV drop over 0° to 70° C was obtained. The power consumption of the bandgap core is $74.35\mu W$. Table 1 outlines all sizing used for this bandgap core design.

The following table outlines all sizing used for this bandgap core design.

TABLE I
SIZING OF COMPONENTS USED IN THE SIMULATED LOW-VOLTAGE BANDGAP REFERENCE CIRCUIT

Component Name	W	L	Value
M1	40u	4u	
T1	20u	10u	
M2, M3	30u	3u	
M4	20u	4u	
M5-M8	16u	12u	

M9, M10	5u	500n	
M11	50u	50n	
M12, M13	32u	1u	
M14	10u	1u	
M15	40u	1u	
C1, C2	19.2u	20u	800 fF
R1	2u	34.15u	64 k Ω
R2	2u	42.8u	55 k Ω
R3	2u	42.8u	5 k Ω
R4, R5	2u	42.8u	60 k Ω
R6	2u	68.7u	40 k Ω
D1-D7	2u	5u	
D0	2u	5u	

B. Start-Up Circuit

To ensure gradual and controlled rise of power during the initialization of our circuit a start-up circuit is used. The start-up circuit used here is adapted from Razvi [1]. The start-up pushes the voltage at the terminal of the op-amp to avoid a time zero ground state. This helps in maintaining stability and reliability during a ramp to V_{DD} .

The start-up circuit is particularly used in our design to control the output terminal of the op-amp to be at an initial voltage, which is determined by the resistor connected to ground, at the start of power-up. With the capacitor uncharged the PFET is able to control the output terminal voltage to be at an initial voltage. As time passes the capacitor is charged therefore PFET is set to a turned-off state. This ensures the start-up circuit is not affecting the bandgap voltage reference in normal operation.

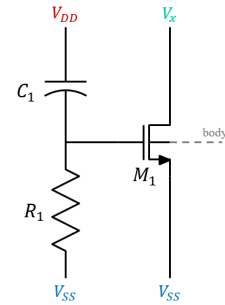


Figure 2: Schematic of the startup circuit, adapted from [1].

This design was chosen because of its simplicity. The circuitry does not have as many components as other start-up circuits, so it makes designing and tuning easier. The resistor sizing was chosen by balancing a low current with a smaller layout. The capacitor sizing was chosen for its small layout size and keeps the PFET in the off state in normal operation.

C. Operational Amplifier

This bandgap reference circuit implements an op-amp to provide a negative feedback control for output voltage adjustment. This feedback aims to provide a high-stability operation condition between the bandgap core and the start-up circuit. To achieve this, the op-amp must achieve a very high gain to resemble the ideal differential amplifier, which will ensure the closed-loop gain is high enough to maintain accuracy.

In this work, a low-voltage, low-power subthreshold op-amp is presented. Operating at 1.2 V, it provides a gain of roughly 55 dB, consuming less than 13.5 μ W. The topology of this circuit is a two-stage Op-amp based on Abdullah-Al-Kaiser design [2]. The first stage is a differential op-amp, where the non-inverting input is connected to a PMOS common source amplifier with Miller capacitance compensation. The bias current is provided by the PTAT generator as well as a current mirror set consisting of two NMOS.

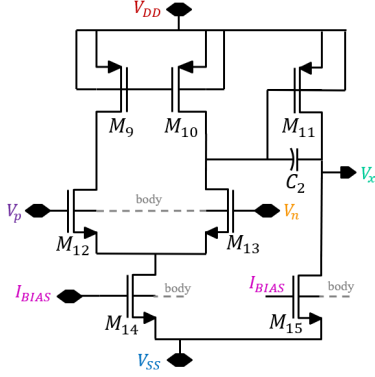


Figure 3: The two-stage op-amp used in the bandgap core, adapted from [2].

In this topology, the gain of the first stage depends on the transconductance of the M_{13} MOSFET and the output resistance of M_{10} and M_{13} , while the gain of the second stage depends on the gain of the M_{11} MOSFET and the output resistance of M_{11} and M_{15} . Aside from sizing, this can also be adjusted by modifying the current mirror and the output of the PTAT generator. The final gain of this op-amp then can be determined by the equation

$$A_V = g_{m,M_{13}}(r_{O,M_{10}} || r_{O,M_{13}}) * g_{m,M_{11}}(r_{O,M_{11}} || r_{O,M_{15}}) [3].$$

Meanwhile, the Miller capacitance here provides better stability to the circuit by moving away the non-dominant pole. One major reason why this topology is embraced in our design is because it allows a decent trade-off between bandwidth and gain. Since this work does not emphasize high-frequency PSRR, the op-amp can prioritize power and gain only. This makes subthreshold operation fit in perfectly, as it requires minimal bias current for very high gain, cutting down the power consumption not only for the op-amp but also for the PTAT generator.

D. PTAT Bias Generator Circuit

To bias the operational amplifier, a PTAT Current Bias Generator is used. The benefit of a current reference is its temperature dependency. Unlike a standard non-ideal current source, which ideally supplies the same current at all temperatures, the resulting current from the bias generator changes with temperature. This helps compensate for the temperature changes that also occur in the op-amp.

The op-amp used in our topology requires a supply current of 1 μ A. Since a small current is needed, both the current mirror and base of the PTAT Current Bias Generator functions in weak inversion (PTAT source). In weak inversion, this module uses low power, 1.2 μ A. Figure 4 showcases the schematic of the

circuit, adapted from Haft-Baradaran [4]. This circuit can be described by the following methodology.

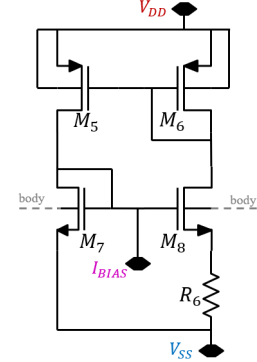


Figure 4: The PTAT Bias Generator for the op-amp, adapted from [3].

Carrier mobility decreases with temperature due to lattice scattering by a factor of $T^{-1.5}$ [5]. The strategy introduced by Haft-Baradaran, which uses a resistor to provide a PTAT voltage source to the MOS source of M_8 , increases with temperature by a factor of $T^{0.5}$. The Mobility and voltage over the resistor R_6 are related by

$$I = \mu_n C_{OX} \left(\frac{W}{L}\right) J_{D0} e^{\frac{V_{M7}}{nV_T}} \left(e^{\frac{-V_{S8}}{V_T}} - e^{\frac{-V_{D8}}{V_T}} \right), \text{ where } V_T \sim 150\text{mV}$$

and $\mu_n C_{OX} \sim 1.83 * 10^{10} \frac{A}{V^2}$ in the transistor models used.

Assuming that $V_{DS,M3}$ is much greater than V_T , Haft-Baradaran further simplifies the equation above to get the equation describing the voltage across the resistor, $V_R = V_T \ln \left(\frac{\beta_5 \beta_8}{\beta_6 \beta_7} \right)$, where $\beta = \mu_n C_{OX} \left(\frac{W}{L}\right)$.

To set the large resistance, an arbitrary value of 50 k Ω was used, which requires a voltage drop across the resistor, $V_R = 0.05\text{V}$, to meet the 1 μ A current requirement of the op-amp. Using the equation describing V_R , this requires the MOSFET dimensions to be very close to equal.

Creating this design, minimum sizing for all transistors were used, while just the resistor was used to create the small current. All the transistors needed to be sized 10x larger due to Monte Carlo component variation.

E. Tradeoffs

The subthreshold region was used in the topology to produce a high gain op-amp and a low current supply reference. The use of the subthreshold region aided in the development of a low voltage design, but with tradeoffs. To bring low variability due to inter-die and intra-die variation, device dimensions, and therefore die area greatly increased to 195.31 μm x 164.42 μm . This design strategy was necessary because many of the devices rode close to the subthreshold region and entered another region with MOS sizing differences [6]. Along with the increase in area, subthreshold designs decrease the speed at which the circuit can run. This increase in delay, and therefore lower operating frequencies, is due to the use of inversion current as drive current [6]. To contribute to this delay, the circuit bandwidth decreased for increased gain in the op-amp. The resulting PSRR was 45dB from low frequency to 50 kHz. After

this band, the op amp may be contributing thermal noise [1], drastically increasing the AC signal passed through the circuit.

Along with these considerations, noise was a further tradeoff for the design of the low voltage topology. One of the benefits of a high voltage system is the large signal to noise ratio. However, as the highest voltage in the circuit decreases, the amplitude of noise voltage that can be tolerated in the system decreases as well to maintain an acceptable signal to noise ratio [7]. The effect of noise was observed when AC transient noise was added to the voltage supply of 1.2V. The addition of noise increased the variance of V_{REF} at the output from 1mV over the transient response to 9mV of cluttered output. A clear output voltage could not be interpreted.

IV. RESULTS

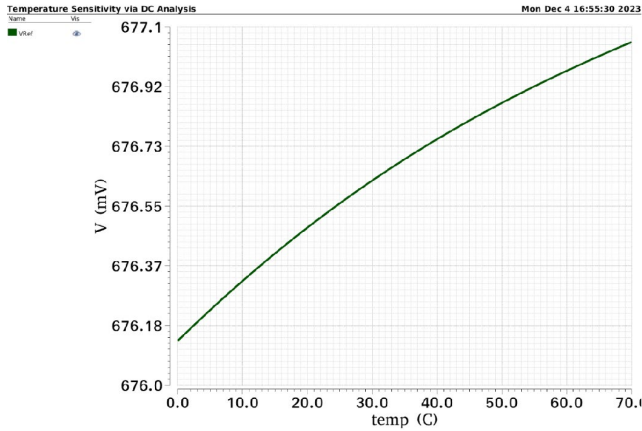


Figure 5: Temperature Sensitivity of the Output Voltage

Figure 5 displays the circuit's stable output voltage over the target temperature range of 0-70°C simulated with a stable DC supply in Spectre. Over the target range the variation of the output voltage is kept minimal within the range of 1mV of overall variation. This result corresponds to a temperature coefficient of 13.14 ppm/°C which is well within the proposed specification of below 50 ppm/°C. Further improvements could be made by tuning resistor values in the core subcircuit to target a subsection of the curve closer to the peak of the output. By centering the voltage peak in the center of the desired temperature range a marginally more stable output can be achieved.

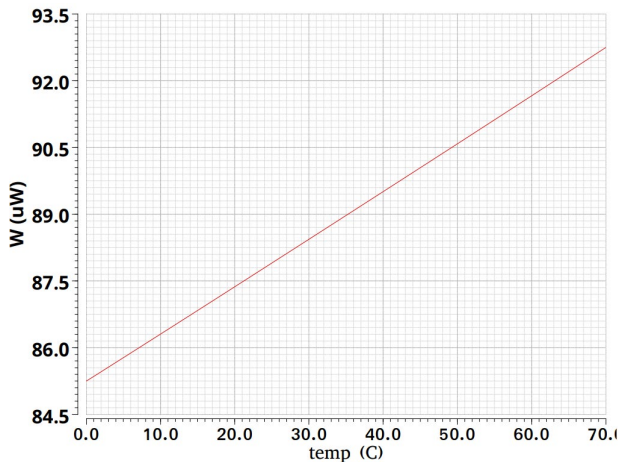


Figure 6: Power Draw over Temperature

Figure 6 displays the circuit's power draw over the target temperature range of 0-70°C simulated with a stable DC supply in Spectre. Over the target range the power never reaches above 100 μ W and can be used in a wide variety of low power applications. This degree of low power is to be expected in this type of circuit as these circuits are widely used and inefficiencies in these circuits could aggregate quickly due to their extensive usage.

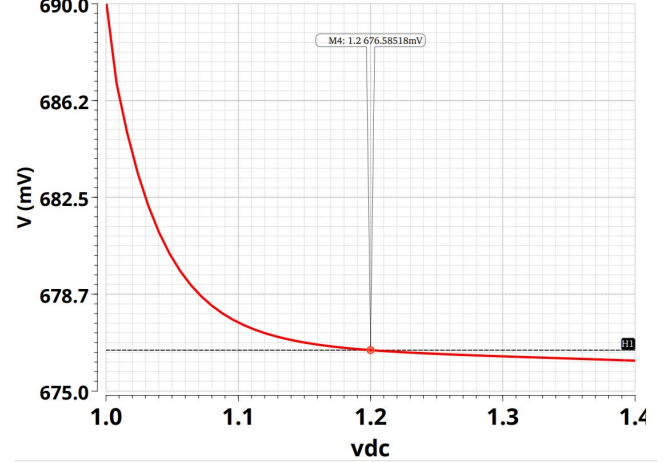


Figure 7: V_{REF} over 20% V_{DD} variation

Figure 7 shows the measured reference voltage of the proposed system under ± 0.2 V power supply variation. As voltage supply increased from 1V to 1.15V, a steep 12mV drop in V_{REF} occurred. For the remaining supply variance from 1.2V to 1.4V, the reference voltage of the output became more stable, and only varied by about 2mV. The system remained in the correct operating regions at both voltage supply variation extremes.

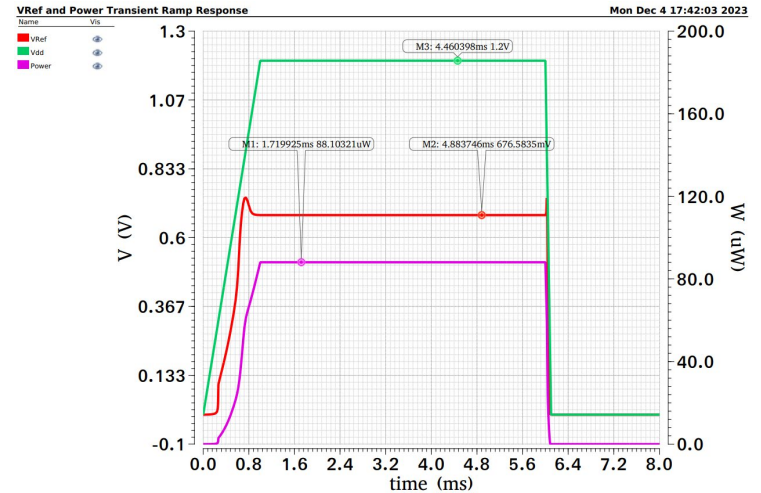


Figure 8: Power Draw over Temperature

Figure 8 depicts the transient performance of the designed bandgap voltage reference. The X-axis is time in milliseconds, whereas the Y-Axis refers to millivolts for V_{REF} and microwatts for power consumption. To initiate our graph we set V_{DD} to ramp from 0V to 1.2V for 1ms, where it then remains constant for 5ms. After, it drops to 0V for 2ms. The ramp time was chosen to 1 millisecond as this is considered a longer ramp time and would properly test our start-up circuit.

The reference voltage steadily increased from 0V to an overshoot of 742mV then back to normal operation voltage of 676.6mV. The quick stabilization and relatively minimal overshoot demonstrates the start-up circuit's robustness. The power consumption of the bandgap reference is also shown. The power consumption is a steady ramp up to 88 μ W. This displays our stabilization and overall low power consumption. This convergent behavior confirms the robustness of our design in effectively managing fluctuations and ensuring smooth turn-on.

V. SUMMARY

Overall, the results of the circuit are satisfactory as the designed specifications meet the proposed benchmarks including below 50 ppm temperature coefficient at 1.2V supply voltage and less than 100 μ A of current draw. These specifications have been verified through transient and DC operation as well as process corners and Monte Carlo variation simulation. Our output voltage remains high to the extent of the 1.2V supply as it is in the upper half of the supplied voltage range. This is not an issue as the output remains highly temperature invariant with comparable output to its peers however a lower voltage output would have been more difficult to support without other more drastic trade-offs. Further sizing considerations for the purpose of greater stability and performance would also improve the overall quality of the circuit. If the resistors and FETs were able to be increased in size with minimal impact to their performance and routing parasitics the sensitivity of this work to mismatch and process variation would further decrease.

Higher yield of the circuit at the trade off of die area is a significant avenue for further improvement. Subsequently the startup circuit is sufficient for the current design but a more robust start up circuit was found in the same paper [1] but required an additional biasing voltage that was deemed too costly with regards to current draw and overall circuit complexity. This more substantial startup circuit used a gain comparison system with an op-amp and inverter to bias an NMOS transistor to the same V_x node of the band gap core. More robust startups like the design just mentioned would be able to be more invariant to process and mismatch variation as they would be better able to perturb the bandgap core into its operational state. This would allow for higher yield and an overall more robust design. Similar improvements to the other component designs like a more efficient op-amp or a more robust current bias would also allow for improvement of the overall circuit.

The overall performance of the proposed design post-PEX followed closely to the performance of Jun's topology. Summarizing the results in Table 2, this work produced a comparable reference with 0.5ppm/ $^{\circ}$ C lower performance at a voltage supply 0.2V less than their work.

TABLE 2
THIS WORK AGAINST COMPARABLE LOW VOLTAGE BANDGAP
REFERENCE CIRCUITS

	This work	[8]	[1]	[9]
Power consumption (μ W)	88.1	162	100	160
Voltage Supply (V)	1.2	1.4	1.0	1.2
Voltage Output (mV)	676	858	500	585
ppm/ $^{\circ}$ C	13.14	12.43	50.00	3.343
Die Area (mm^2)	0.32	1.2	N/A	0.00085
Operating Temperature	0-70	N/A	0-100	0-125

VI. REFERENCES

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